

Synopsis

House Listing Model

Problem Id: 5

Student Name: Ashu Khanduri

University Roll number: 2028673

Section: K

Student Name: Srishti Nahar

University Roll number: 2028187

Section: K

Student Name: Anushka Pun

University Roll number: 2028656

Section: K

1. Problem Description:

Design and develop a regression-based machine learning system to predict house sale price using a structured real-estate dataset consisting of 25 property-related features. The dataset represents residential housing information collected from different regions and includes both numerical and categorical attributes that influence property valuation.

The dataset contains the following 25 input features: property area (square feet), number of bedrooms, number of bathrooms, number of floors, year built, property age, renovation status, lot size, distance to city center (km), neighborhood quality score, crime rate index, nearby school rating, hospital proximity (km), shopping center proximity (km), public transport accessibility score, parking availability (binary), property type (categorical), construction quality rating, energy efficiency score, water supply reliability score, electricity supply reliability score, internet availability score, green space availability index, flood risk index, and noise pollution level.

The project should begin with Exploratory Data Analysis (EDA) to analyze feature distributions, correlations, and their relationship with the target variable. Perform data preprocessing including handling missing values, encoding categorical features, and normalization or feature scaling.

Develop and train suitable regression models to predict house prices and visualize predicted versus actual values as well as residuals. Finally, evaluate the trained models using appropriate regression performance metrics such as R² score and error measures such as MAE, MSE, and analyze the impact of different features on the prediction outcome. The project should demonstrate a complete machine learning pipeline from raw data to model evaluation and interpretation.

1.1. Dataset Description

Name: Real_House_Data.csv Task Type: Regression (Supervised Learning) Target Variable: Price — continuous numeric value representing the listed house price.

Features Overview

The dataset contains a mix of structural, locational, and environmental attributes of residential properties.

Structural Features

- Property_Area: total area of the property (continuous)
- Year_Built : year of construction (later dropped as a direct feature)
- Property_Age: derived age of the property
- Bedrooms: number of bedrooms
- Lot_Size: size of the plot; contained missing values imputed by property type and bedroom group mean
- Parking_Availability: binary (0 = No, 1 = Yes)
- Construction_Quality_Rating: quality score on a scale of 1–10
- Renovation_Status: categorical (Yes / No), one-hot encoded

Locational Features

- Distance_to_City_Center — distance in km from the city center

- Neighborhood_Quality_Score — neighbourhood rating on a scale of 1–10
- Nearby_School_Rating — rating of nearest school; imputed using neighbourhood quality bins
- Crime_Rate_Index — area-level crime index (continuous)
- Green_Space_Availability_Index — proximity/availability of green spaces; imputed by distance bins

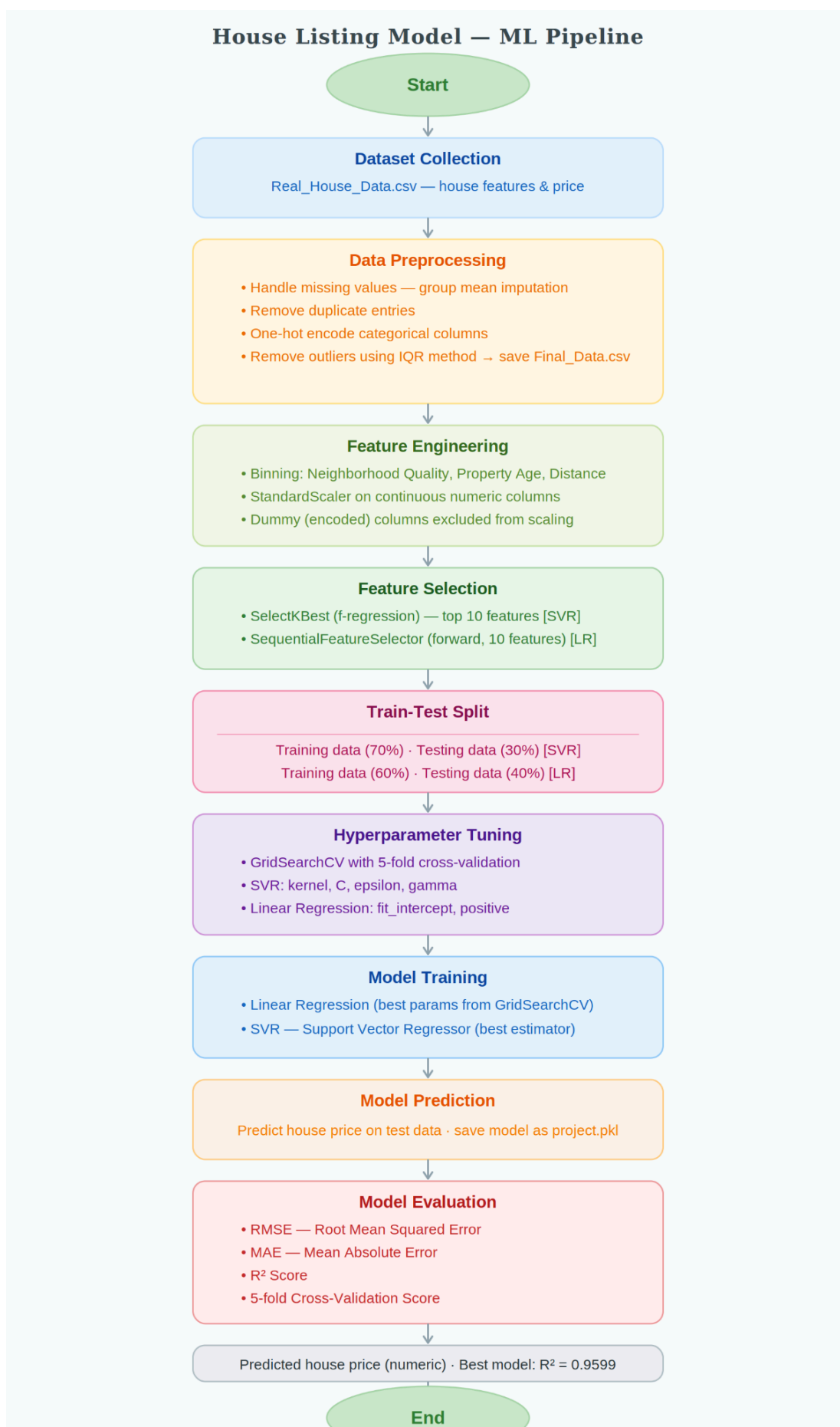
Environmental Features

- Flood_Risk_Index — flood risk score for the property location
- Energy_Efficiency_Score — energy performance rating; imputed using property age and construction quality groups

Property Type

- Property_Type — categorical with three classes: Apartment, Villa, Independent House; one-hot encoded (drop-first applied)

1.2. Flow Diagram



2. Models for Project Implementation

1.1. Models Used

- Linear Regression: The baseline model. It fits a straight-line relationship between selected features and house price by minimizing the sum of squared residuals. Simple, fast, and highly interpretable coefficient values directly show how much each feature influences price. Works best when relationships are approximately linear and multicollinearity is low.
- Random Forest Regressor: An ensemble of decision trees, each trained on a random subset of data and features. Predictions are averaged across all trees, which reduces variance and handles non-linear relationships naturally. It is robust to outliers and requires minimal feature scaling, making it a strong performer on tabular real-estate data.
- ElasticNet: A regularized linear model combining L1 (Lasso) and L2 (Ridge) penalties. L1 shrinks less important feature coefficients to zero (automatic feature selection), while L2 handles correlated features gracefully. Particularly useful here because house features like area, rooms, and quality scores tend to be correlated and ElasticNet prevents any single correlated feature from dominating the model.

1.2. Platform Used

- Jupyter Notebook

3. Machine Learning Model/Searching Technique used:

1. Linear Regression

Type: Parametric Supervised Learning (Regression)

The baseline model. It estimates house price as a weighted linear combination of the selected input features by minimizing the Residual Sum of Squares (RSS). Coefficients are directly interpretable, each value represents the marginal change in price per unit change in that feature, holding others constant. Hyperparameters (`fit_intercept`, `positive`) were tuned using GridSearchCV.

2. Random Forest Regressor

Type: Ensemble Learning (Bagging)

An ensemble of independently trained decision trees, each built on a random bootstrap sample of the data and a random subset of features. Final predictions are obtained by averaging outputs across all trees, which significantly reduces variance without increasing bias. Naturally handles

non-linear feature interactions and is robust to outliers in the training data.

3. ElasticNet

Type: Regularized Linear Regression (L1 + L2 Penalty)

A hybrid regularization model combining Lasso (L1) and Ridge (L2) penalties. The L1 component performs automatic feature selection by shrinking less important coefficients to exactly zero, while the L2 component handles multicollinearity by distributing weight across correlated features. Particularly effective here given the expected correlations among features like area, quality ratings, and price.

4. Support Vector Regressor (SVR)

Type: Kernel-Based Supervised Learning (Regression)

An adaptation of Support Vector Machines for continuous output prediction. SVR fits a function within a defined tolerance margin (epsilon-tube) while minimizing model complexity, making it resistant to minor noise in the data. The RBF (Radial Basis Function) kernel was applied to capture non-linear relationships between features and price. Achieved the best recorded R^2 of 0.9599.

Searching / Optimization Techniques

GridSearchCV: An exhaustive hyperparameter search strategy that evaluates every combination of a defined parameter grid. Applied to both SVR and Linear Regression to identify the optimal model configuration. Integrated with 5-fold cross-validation to ensure that selected parameters generalize well beyond the training set.

Model	Parameters Searched
SVR	kernel, C, epsilon, gamma
Linear Regression	fit_intercept, positive

SequentialFeatureSelector (Forward Selection): A wrapper-based feature selection method used with Linear Regression. It iteratively adds the single most beneficial feature at each step based on cross-validated model performance until the desired count of 10 features is reached. This

ensures only the most predictive subset of features enters the final model.

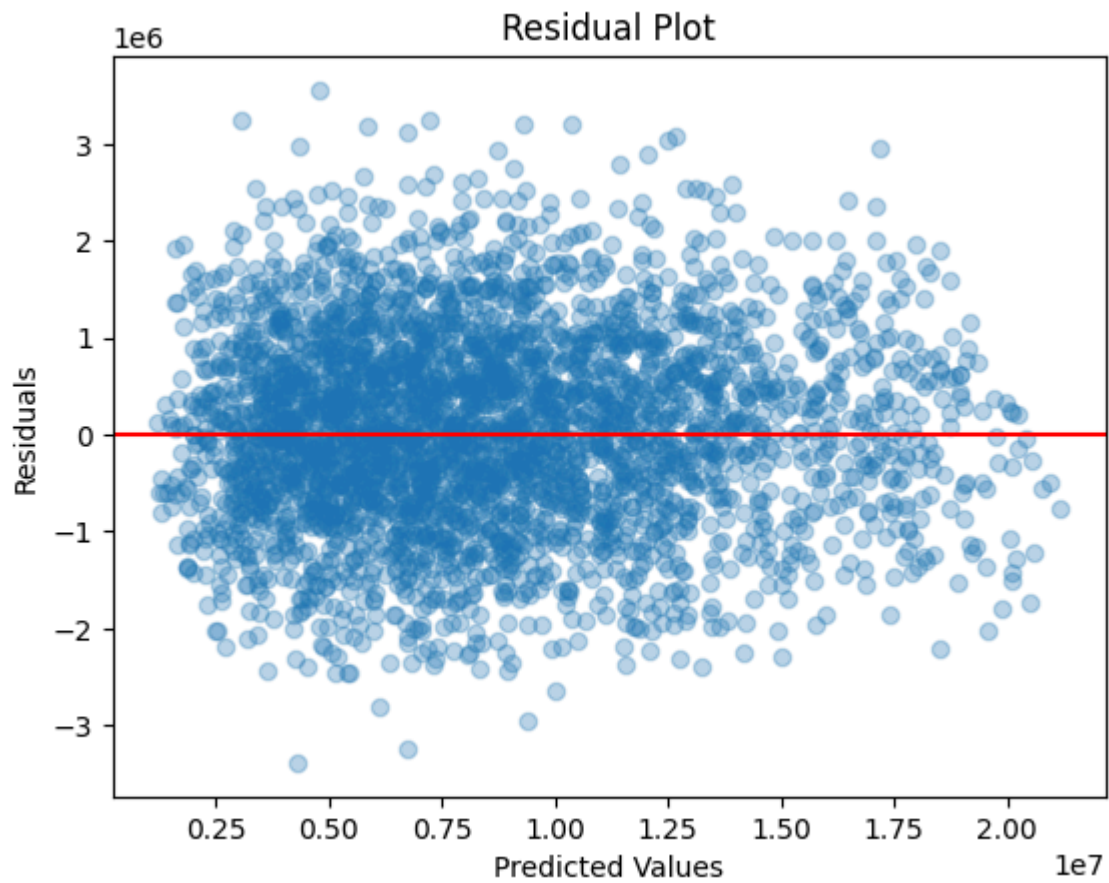
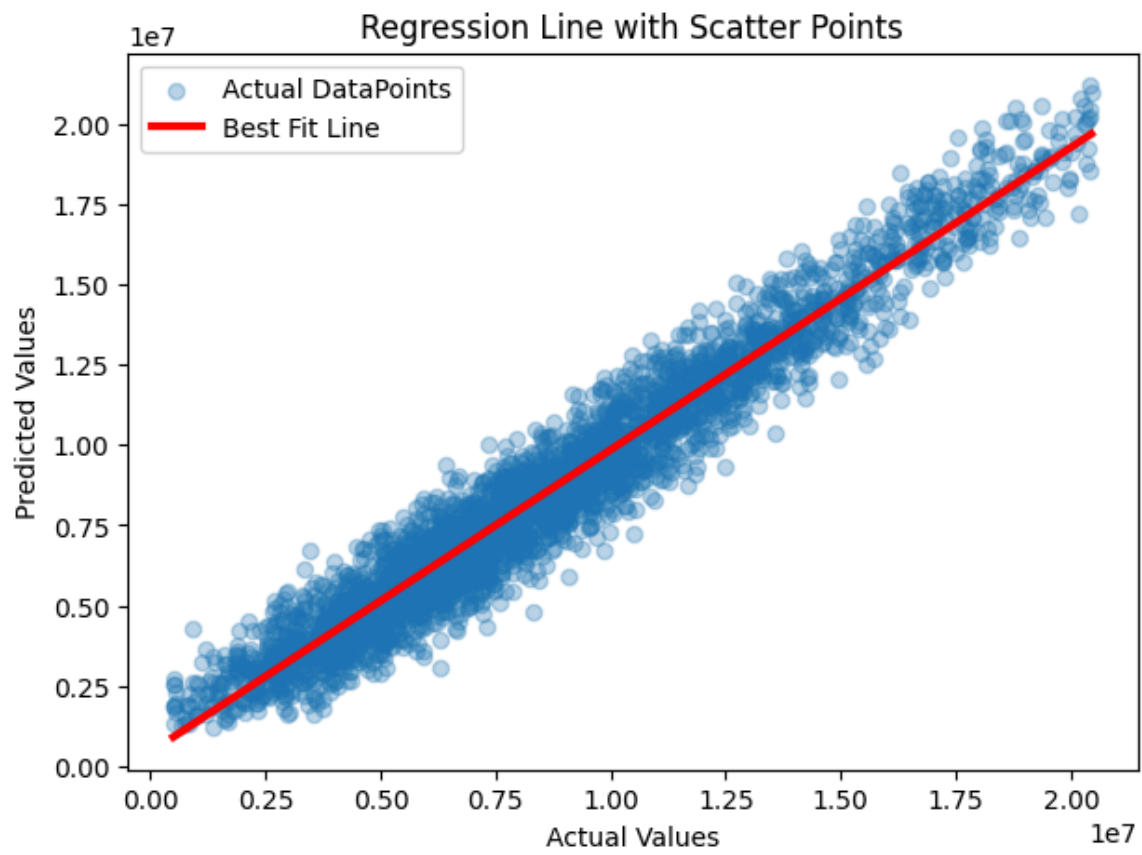
SelectKBest (f-regression) A filter-based feature selection technique used with SVR. It scores each feature independently using the F-statistic of its linear correlation with the target variable and retains the top K (10) highest-scoring features. Fast, statistically grounded, and effective as a preprocessing step before kernel-based models.

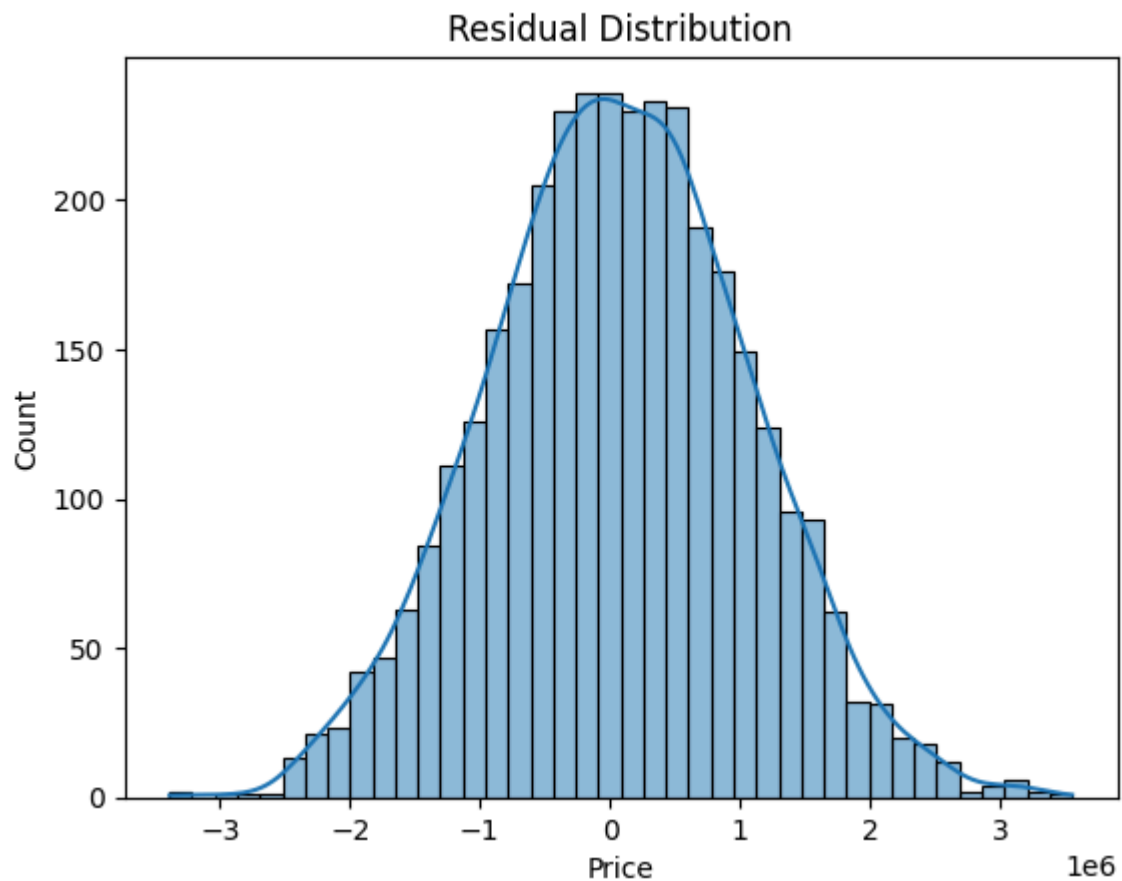
4. Expected Evaluation metrics and Plots to be used:

- RMSE (Root Mean Squared Error): It measures how far off the predictions are on average, in the same unit as price. Lower = better. It punishes big errors more than small ones.
- MAE (Mean Absolute Error): It is the average of how much the predictions are off (e.g., "off by ₹5 lakh on average"). It's easier to interpret than RMSE.
- R² Score: It tells how well the model explains price variation. $R^2 = 1.0$ is perfect; $R^2 = 0.85$ means the model explains 85% of why prices vary. Higher = better.
- 5-fold Cross-Validation: It splits data into 5 chunks, trains 5 times, and averages R² across all runs and checks that the model isn't just memorizing training data (overfitting).

5. Results & Visualizations

- Price Distribution (Histogram + KDE): Reveals the statistical spread of target values. A right-skewed distribution (common in real estate) indicates that a few high-value properties dominate the range, justifying outlier removal prior to modeling.
- Scatter Plots (Area / Distance / Crime vs Price): Exploratory analysis confirming expected feature-price relationships: larger property area correlates positively with price, while greater distance from the city center and higher crime index correlate negatively. This validates the relevance of these features before model training.
- Boxplot of Price: Identifies statistical outliers using the IQR method. Extreme values are flagged and removed to prevent them from distorting regression coefficients and inflating error metrics.
- Actual vs Predicted Scatter (with Best Fit Line): A key diagnostic post-training. Points clustering tightly along the diagonal best-fit line indicate low prediction error. Systematic deviation suggests model bias or unaccounted non-linearity.
- Residual Plot (Predicted vs Residuals): Tests a core linear regression assumption that errors are randomly and evenly distributed. A fan-shaped or curved pattern would indicate heteroscedasticity or a need for non-linear modeling.
- Residual Distribution (Histogram + KDE): Validates the normality of errors, a prerequisite for reliable inference in linear models. A bell-shaped curve centered at zero confirms well-calibrated predictions.





```
print("\n Predicted Price:", prediction[0])
```

Model R2 Score: 0.9599320529599936

Property Area: 2684.82

Year Built: 2018

Distance to City Center: 33.35

Neighborhood Quality Score (1-10): 8

Parking (0 or 1): 1

Construction Quality (1-10): 2

Flood Risk Index: 42.26

Renovation Status (Yes/No): yes

Property Type (Apartment/Villa/Independent House): independent house

 Predicted Price: 11652556.62895305

6. Conclusion and Future Scope

Conclusion

The House Listing Price Prediction model successfully applies supervised machine learning to estimate residential property prices based on structural, locational, and environmental attributes. Across four models evaluated; Linear Regression, Random Forest Regressor, ElasticNet, and SVR, the Support Vector Regressor delivered the best performance with an R^2 score of 0.9599. The combination of rigorous preprocessing, targeted feature selection, and GridSearchCV-based hyperparameter tuning resulted in a reliable and well-generalized price prediction pipeline.

Future Scope

The model can be further enhanced by incorporating additional features such as proximity to transport, hospital ratings, and live market trends. Advanced algorithms like XGBoost or LightGBM could be explored to improve predictive accuracy. Deploying the model as a real-time web application would make it practically accessible to buyers, sellers, and real estate professionals. Training on larger, geographically diverse datasets and periodic retraining with updated market data would further strengthen its scalability and relevance.

7. References

- Books and Link Sources:

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- Gemini/Claude AI : For helping in understanding what to do and why we are doing it.
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- Scikit-learn Official Documentation- <https://scikit-learn.org/docs/>
- Geeks for Geeks website: <https://www.geeksforgeeks.org/>